MAHEDI HASAN

+1-509-715-8554 | mhasan32@wisc.edu | Madison, WI

mdmahedihasan-statistics | mahedihasanstat | mahedihasanstat.github.io

Current Affiliation

- **Postdoctoral Research Associate** *2025 – Current*

  Dept. of Statistics, University of Wisconsin–Madison
  - Advised by Dr. Jun Zhu (Dept. of Statistics, UW–Madison) and Dr. Dan Walsh (U.S. Geological Survey / University of Montana).
  - Developing statistical frameworks for modeling infectious disease transmission in dynamic, heterogeneous wildlife systems.
  - Developing frameworks that integrate MCMC and Bayesian inference, ODE/PDE models, and spatio-temporal graph neural networks, embeddings, and uncertainty quantification.

Research Interests

I work at the intersection of statistics and machine learning to build methods and tools that connect theory with application, solving complex real-world research problems with trustworthy, defensible solutions.

Statistical Methods: Bayesian Inference, Uncertainty Quantification, Spatial Statistics, Experimental Design, Mixed Models & Longitudinal Data Analysis, Multivariate Methods, Time Series Models, Change Point Detection, Optimization, Bioinformatics, Network Modeling.

Machine Learning & AI: Deep Learning, Generative AI, Graph Neural Networks, Interpretable ML, Computer Vision.

Education

- **Ph.D. in Statistical Science** *2019 - 2025*
  Dept. of Mathematics and Statistics, Washington State University (WSU), Pullman, WA
  Advisor: Dr. Daryl DeFord
  Dissertation: Inference, Aggregation and Embedding for Learning Problems on Network and Time Series Data

- **M.S. in Computer Science** *2021 - 2023*
  EECS, Washington State University (WSU), Pullman, WA
  Specialization: Machine Learning and AI

- **Graduate Certificate in Bioinformatics** *2022 - 2024*
  School of Biological Sciences, Washington State University (WSU), Pullman, WA
  Concentration: sequencing algorithms, omics data analysis, biological network analysis.

- **Master of Philosophy (M.Phil.) in Applied Statistics** *2017 - 2019*
  Bangladesh University of Professionals (BUP), Dhaka, Bangladesh
  Thesis: Impact of Climate Change on Agricultural Production in Bangladesh
  Mentor: Dr. Syed Shahadat Hossain, ISRT, University of Dhaka.

- **M.S. in Statistics** *2011 - 2012*
  Dept. of Statistics, Shahjalal University of Science and Technology (SUST), Sylhet, BD.
  Ranked 1st in cohort; Distinction (CGPA: 3.95/4.00)
  Thesis: Regional Variation of Climate Change in Bangladesh: Estimation of Missing Values, Extremity & Trend
  Mentor: Dr. Md Zakir Hossain, SUST.

- **B.Sc. in Statistics** *2006 - 2010*
  Dept. of Statistics, Shahjalal University of Science and Technology (SUST), Sylhet, BD.
  Ranked 1st in cohort; Distinction (CGPA: 3.91/4.00)

Honors and Awards

- **Winner of "2024 Data Challenge Expo" at JSM** *2024*
  Joint Statistical Meeting (JSM), Portland Oregon. American Statistical Association (ASA)
  ◦ Secured third place in the nationwide General Social Survey (GSS) data competition.
- **President's Award for Leadership at WSU** *2023*
  Washington State University
  ◦ For exemplary leadership and community engagement at WSU.
- **Teaching Excellence Award as Graduate Student Instructor** *2022*
  Graduate and Professional Students Association, Washington State University
  ◦ Acknowledged for outstanding teaching performance as a graduate student instructor at WSU.
- **Three Minutes Thesis Contest Winner** *2022*
  Dept. of Mathematics and Statistics, Washington State University
  ◦ Secured second place in the department-level competition and advanced to the college-level competition.
- **Sidney G. Hacker Mathematics Scholarship** *2022*
  Dept. of Mathematics and Statistics, Washington State University
  ◦ For exceptional community engagement and service, Washington State University.
- **Best Research Paper Award** *2018*
  International Conference Organized by AMDISA, Uttara University, Dhaka, Bangladesh
  ◦ Selected as the best paper among over 70 papers presented in the conference.
- **Academic Distinction Award** *2010 & 2012*
  Shahjalal University of Science and Technology, Sylhet, Bangladesh.
  ◦ For achieving the top position with the highest CGPA in both undergraduate and Master's programs.

Teaching

- **University of Wisconsin–Madison — Lecturer** *Spring 2026*
  Nelson Institute for Environmental Studies
  ◦ STAT 679: Spatial Statistics for Lattice Data *(graduate)*
- **Washington State University — Graduate Student Instructor** *2019 – 2022*
  Dept. of Mathematics and Statistics
  ◦ STAT 419: Introduction to Multivariate Statistics *(upper-division undergraduate)* Fall 2022
  ◦ Math 216: Discrete Mathematics *(undergraduate)* Spring 2022
  ◦ STAT 212: Introduction to Statistical Methods *(undergraduate)* Fall 2021, Spring 2020
  ◦ Math 172: Engineering Calculus II *(undergraduate)* Spring 2021
  ◦ Math 171: Calculus I *(undergraduate; teaching assistant, lab sections)* Fall 2019
  Taught 150+ students across disciplines and academic levels; also tutored undergraduates at the Math Learning Center; recipient of the GPSA Teaching Excellence Award as Graduate Student Instructor.
- **Bangladesh University of Professionals — Assistant Professor; Lecturer** *2015 – 2019*
  Faculty of Business Studies
  ◦ ECON-4802: Econometrics *(graduate / undergraduate)* Spring 2018, 2019; Fall 2016
  ◦ BUS/ALD-3610: Business Research Methodology *(graduate / undergraduate)* Fall 2016 – 2018
  ◦ ECON-4703: Software Application for Economic Analysis *(undergraduate)* Spring 2018
  ◦ ALD-3509: Business Statistics *(undergraduate)* Spring 2016 – 2019
  ◦ Statistics for Development Studies *(undergraduate)* Fall 2017
  ◦ ALD-1205: Business Mathematics *(undergraduate)* Fall 2017
  ◦ ECON-104: Statistics for Economics I *(undergraduate)* Fall 2015
  Taught 900+ students; supervised 14 research students through thesis completion.
- **National Defence College (NDC) — Guest Lecturer** *2018 – 2019*
  Mirpur, Dhaka, Bangladesh
  ◦ Held a concurrent guest faculty position, providing hands-on training in research methodology and scientific report writing to civil and military personnel.

Published Works

[1] Hasan, M. M., Wooyoung K., Jacqueline C., and David R. (2025). **Investigating American Optimism: Evidence from 2022 GSS Data.** CHANCE 38, no. 2: 44-51. https://doi.org/10.1080/09332480.2025.2510164

[2] Ahmed, M., Ahamad, M.G., **Hasan, M. M.** (2025). **Google search interests and new cases of COVID-19 in Bangladesh: a vector autoregression analysis for disease surveillance.** BMC Res Notes 18, 312. https://doi.org/10.1186/s13104-025-07381-2

[3] Kimberly L. M., Kathryn E. S., Elizabeth A. J., Andrew J. S., Anya L. S. , Kathryn G., Riley T., **Hasan, M. M.** & Marc G. B. (2024) **Evidence for environmental influences on impulsivity and aggression** . in *Urban Forestry & Urban Greening, 103*.

[4] Anjum, A., Ahammed, T., **Hasan, M. M.**, Chowdhury, M. A. B., & Uddin, M. J. (2023). **Mothers' functional difficulty is affecting the child functioning: Findings from a nationally representative MICS 2019 cross-sectional survey in Bangladesh**. In *Health Science Reports 6(1)*.

[5] Shahriar, S., Qian, L., Rahman, A., **Hasan, M.**, Kea, S., & Abdullahi, N. M. (2020). **Youth skill development loans (YSDL) and good governance in Bangladesh: a logit model analysis**. *Emerging Markets Finance and Trade*, 56(11), 2529-2542. https://doi.org/10.1080/1540496X.2019.1594769

[6] Nury, A. H., Hasan, K., Alam, M. J. B., & **Hasan, M. M** (2017). **Analysis of Time Series Variations of Temperature and its Forecast in Northeastern Bangladesh**. *International Journal of Global Warming*, 13(2), 157-182. https://doi.org/10.1504/IJGW.2017.086283

[7] **Hasan, M. M.** , Islam, M. T., Sultana, N. , &(2017). **Effect of Climate Change on Rice Production in Bangladesh: A Comparative Study between Time Series and Stochastic Models**. *BUP Journal*, 4 (2).

[8] Sultana, N., Islam, M. T., & **Hasan, M. M.** (2017). **An Inter-Services Comparative Study on the Diversification of the Factors of Employee Retention in Bangladesh: Application of Structural Equation Modeling**. *Australian Academy of Business and Economics Review*, 3 (3), 166-176.

[9] Ahmed, M. U., Muzib, M. M., & **Hasan, M. M.** (2016). **Inflation, Inflation Uncertainty and Relative Price Variability in Bangladesh**. *Eurasian Economic Review*, 6 (3), 389–427. https://doi.org/10.1007/s40822-016-0055-8

Work in Progress (in-progress research titles are tentative)

[1] **Improving Generalizability of Deep Learning Models for Discriminating Earthquakes and Explosions**. [**Current Status:** Work-in-Progress]

[2] **Interpretable Deep Learning Model for Discriminating Earthquakes and Explosions**. [**Current Status:** Work-in-Progress]

[3] **Change Point Detection for Dimensionality in Random Dot Product Graphs**. **Current Status:** Work-in-Progress.

[4] **Studying Social Network Dynamics: Addressing Aggregation Challenges and Modeling Language Risk in School Friendship Networks. Current Status:** Work-in-progress.

[5] Smith, C. L., Avery, A., Ryan, R., Duncan, G. E., Colletto, S., **Hasan, M. M.**, Hall, L., McPherson, S. M. **Genetic and environmental influences on the relationships between smoking and cannabis co-use and their association with chronic pain: A twin study**. [**Current Status:** Ready-to-Submit]

Conference Presentations

[1] **Hasan, M. M.** (2026). Latent Representations of Dynamic Contact Networks in Wildlife Health Modeling and Uncertainty Quantification. *Emerging Directions in Spatial Statistics (ENVR)*. University of Houston, Houston, TX, USA, Oct 8–10 *(scheduled)*.

[2] **Hasan, M. M.** (2026). Movement-Derived Contact Networks and Latent Embeddings for Wildlife Health Modeling and Uncertainty Quantification. *SIAM Conference on Uncertainty Quantification*. Minneapolis, MN, USA.

[3] **Hasan, M. M.** (2025). Embedding-Based Animal Trajectory Prediction Using Representation Learning. *ISU-NISS Conference on AI and Statistics*. Iowa State University, Ames, IA, USA.

[4] **Hasan, M. M.** & DeFord, D. (2024). Studying Social Network Dynamics: Addressing Aggregation Challenges and Modeling Language Risk. *Joint Statistical Meeting*. Portland, Oregon, USA.

[5] **Hasan, M. M.**, Kim. W., Carlton-Wargo, J. & Rice, D. (2024). Investigating American Optimism: Evidence from 2022 General Social Survey (GSS) Data. *Joint Statistical Meeting*. Portland, Oregon, USA.

[6] **Hasan, M. M.** & DeFord, D. (2024). Social Network Dynamics: Addressing Aggregation Challenges in School Friendship Networks. *SIAM Annual Meeting*. Spokane, WA, USA.

[7] **Hasan, M. M.** (2017). Climate Change and Its Impact on Agricultural Production in Bangladesh: Geo-statistical Analysis of Spatial and Temporal Variation. *9th Biennial Conference of Indian Society for Ecological Economics (INSEE)*. Kerala, India.

[8] **Hasan, M. M.** & Hossain, M. Z. (2017). Climate Change in Terms of Extreme Weather Events in Bangladesh: Impacts on the Economic Development. *16th Biennial Conference of Bangladesh Statistical Association*. Dhaka, Bangladesh.

[9] **Hasan, M. M.** & Hossain, M. Z. (2016). Engendering Extremity in the Climate of Bangladesh: An Observation through Maximum and Minimum Temperature. *National Conference on Sustainable Development*. SUST, Sylhet, Bangladesh.

[10] **Hasan, M. M.** & Sultana, N. (2016). An Inter-Services Comparative Study on Job Satisfaction in Bangladesh: Application of Discriminant Function Analysis and Structural Equation Modeling. *International Conference on Business Analytics and Big Data*. Indian Institute of Management (IIM), Ahmadabad, India.

[11] **Hasan, M. M.** & Sultana, N. (2016). An Inter-Services Comparative Study on the Diversification of the Factors of Employee Retention in Bangladesh: Application of Structural Equation Modeling. *International Conference organized by AMDISA*. Uttara University, Dhaka, Bangladesh.

[12] **Hasan, M. M.** & Hossain, M. Z. (2014). Regional Variation of Climate Change through Drought and Trend Analysis using Temperature and Rainfall Data. *Bi-annual Conference of Bangladesh Statistical Association (BSA)*. Dhaka, Bangladesh.

[13] **Hasan, M. M.** & Hossain, M. Z. (2014). Rainfall Variation in Bangladesh: A Wavelet Transformation Analysis. *Bi-annual Conference of Bangladesh Statistical Association (BSA)*. Dhaka, Bangladesh.

Funded Research Projects

- **Time Series Analysis of the Economic Factors of Development in Bangladesh** *2017 - 2018*
  Position: Project Lead, Funded By: Center for Higher Studies and Research, BUP
  - Led the development of the research proposal, study design, survey creation and execution, data analysis, and report preparation.
- **Factors and Strategies for Effective Employee Retention in the Organizations of Bangladesh** *2017 - 2018*
  Position: Project Lead, Funded By: University Grants Commission (UGC), Bangladesh.
  - Led the development of the research proposal, study design, survey creation and execution, data analysis, and report preparation.
- **Rana Plaza Disaster and the Aftermaths: Vulnerability Assessment of the Victims' Children** *2016 - 2017*
  Position: Project Lead, Funded By: University Grants Commission (UGC), Bangladesh.
  - Led the research team in writing a research proposal, designing the study, designing the survey for data collection, data analysis, and report writing.
- **Regional Variation of Climate Change: an Application of Wavelet Transformation** *2013 - 2014*
  Position: Research Assistant, Funded by: Research Center, SUST, BD
  - Proposal writing, Data analysis, and report preparation.

Research Experience

- **Los Alamos National Laboratory (LANL), Los Alamos, NM** *Jun 2024 – July 2025*
  Graduate Research Associate
  - Built CNN, BiLSTM and hybrid models for earthquake–explosion discrimination across synthetic and real seismic domains, improving cross-domain robustness and generalization using adversarial training and transfer learning.
  - Used Grad-CAM for interpretability, showing how models made decisions based on physically meaningful signal regions.
- **Analytics and Psychopharmacology Laboratory (APPL), College of Medicine, WSU** *May 2023 – May 2024*
  Research Assistant / Statistician
  - Statistician in a 15-member research group; guided method selection and analysis for substance-use clinical trials.
  - Prepared complex, multi-visit participant data collected via REDCap for statistical modeling.
- **Center for Interdisciplinary Statistical Education and Research (CISER), WSU** *Aug 2022 – May 2023*
  Statistical Consultant / Research Assistant
  - Advised 30+ interdisciplinary research projects on study design, model selection and interpretation of results.
  - Delivered analyses in each client's preferred software (R, Python, SPSS), working with researchers across disciplines.

Technical Skills

Languages: Python, R, SQL, MATLAB, SAS, SPSS, STATA
ML & DL: PyTorch, TensorFlow, scikit-learn, PyG, DGL; CNNs, LSTMs, GNNs, Transformers
Computing: Linux, Git, Docker, AWS, PySpark, HTCondor, GPU clusters
Statistical Methods: Bayesian inference (MCMC), causal inference, uncertainty quantification, conformal prediction, spatial statistics, experimental design, time series, longitudinal & network models
Data & Reproducibility: REDCap, reproducible pipelines, LaTeX

Service and Professional Activity

- **President, Student Chapter of the American Statistical Association** *2022 – 2023*
  Washington State University
- **Senator, Graduate and Professional Students Association (GPSA)** *2021 – 2022*
  Representing the College, Washington State University
- **Data-literacy outreach** *2022 – 2024*
  Five workshops delivered in schools across Washington State
- **Invited data science workshops** *2023*
  Delivered in R for industry professionals